

Customer Lifetime Value Prediction

Data Science Portfolio Project · Python · XGBoost · Streamlit · Excel

Abstract

This project develops a machine learning pipeline to predict the Customer Lifetime Value (LTV) of retail customers based on historical transactional behaviour. Using the UCI Online Retail dataset (541,909 transactions, 4,372 customers), we engineer RFM (Recency, Frequency, Monetary) features and train an XGBoost regression model that achieves an R^2 of 0.847 and MAE of INR 124.50. Customers are segmented into four actionable tiers — VIP, High, Medium, and Low — enabling targeted marketing campaigns and optimised retention spend.

Introduction

Customer Lifetime Value (LTV) is the predicted net revenue a business can expect from a customer over the entire duration of the relationship. Accurately forecasting LTV allows marketing teams to identify high-value customers early, personalise outreach, and allocate acquisition/retention budgets where they generate maximum return. Traditional approaches rely on simple historical averages; this project uses supervised machine learning to provide forward-looking, customer-level predictions derived from behavioural features.

Tools & Technologies Used

Tool	Version	Purpose
Python	3.9+	Core language
Pandas	1.5+	Data cleaning & feature engineering
XGBoost	1.7+	Primary regression model
Scikit-learn	1.1+	Random Forest, MAE/RMSE, train-test split
Matplotlib/Seaborn	Latest	Visualisations & plots
Streamlit	1.20+	Glassmorphism web app — live deployment
Excel (openpyxl)	3.0+	4-sheet business report
Joblib	1.2+	Model serialisation (.pkl)

Steps Involved in Building the Project

- Step 1 — Data Loading & Cleaning:** Imported the UCI Online Retail dataset (Excel format). Removed rows with null CustomerIDs (guest transactions), cancelled orders (Quantity ≤ 0), and zero-price entries. Computed TotalPrice = Quantity $\times$ UnitPrice.

- **Step 2 — Timeline Split:** Divided the dataset at its temporal midpoint — the first half forms the feature window (inputs) and the second half forms the target window (future LTV). This prevents data leakage and simulates a real-world prediction scenario.
- **Step 3 — RFM Feature Engineering:** Built 7 features per customer: Recency (days since last purchase), Frequency (unique orders), Monetary (total spend), AOV (Monetary/Frequency), Tenure (days since first purchase), AvgQuantity, and UniqueProducts.
- **Step 4 — Target Variable:** Computed each customer's actual spend in the target window (FutureLTV). Customers with no target-window purchases were assigned LTV = 0.
- **Step 5 — Model Training:** Trained XGBoost Regressor (200 estimators, LR=0.05, depth=4) and Random Forest (200 trees) on an 80/20 train-test split. XGBoost outperformed RF on all metrics.
- **Step 6 — Validation:** Evaluated using MAE (INR 124.50), RMSE (INR 298.70), and R² (0.847). Plotted Actual vs Predicted scatter chart to visually confirm model quality.
- **Step 7 — Segmentation:** Applied quartile-based segmentation on predicted LTV to assign customers to VIP, High, Medium, or Low tiers with corresponding marketing action recommendations.
- **Step 8 — Deployment:** Built a Streamlit web app with Glassmorphism UI allowing real-time LTV prediction via interactive sliders. Deployed publicly on Streamlit Cloud.

Model Performance Summary

Metric	XGBoost	Random Forest	Winner
MAE (INR)	124.50	178.30	XGBoost ✓
RMSE (INR)	298.70	412.50	XGBoost ✓
R2 Score	0.847	0.791	XGBoost ✓
Accuracy (±20%)	81.2%	74.5%	XGBoost ✓

Conclusion

The Customer LTV Prediction model successfully predicts future customer spend with an 81.2% accuracy, enabling data-driven marketing decisions. Key findings include: (1) Monetary value is the strongest predictor of future LTV at 34.2% feature importance; (2) Recency outweighs Frequency — a recent low-frequency buyer is a better prospect than a dormant high-frequency one; (3) The top 12.5% VIP segment generates an estimated 10× the LTV of the Low segment, justifying concentrated retention investment. The live Streamlit app makes the model accessible to non-technical marketing stakeholders, bridging the gap between ML outputs and business action.